

Dominic Nguyen

Vancouver, BC

(604)-716-3930 | DominicNguyen817@gmail.com | www.linkedin.com/in/dominicn17

TECHNICAL SKILLS

Computer: SOLIDWORKS, C, MATLAB, Microsoft Excel, KiCAD, Python

Laboratory: Soldering, Oscilloscope, Digital Multimeter, Circuit Analysis, 3D Printing

Machine Tools: Bandsaw, Milling Machine, Lathe, Drill Press, Press Brake

TECHNICAL PROJECTS: [PROJECT PORTFOLIO](#)

Biped Wheeled Robot Pair Project, UBC

December 2025 – February 2026

- Designed a cascaded PID control system for real-time robot balance and speed control, tuning PID parameters using a MATLAB simulation.
- Implemented sensor fusion using a Madgwick filter by tuning data weightings from an accelerometer and a gyroscope, allowing accurate real-time tilt estimation for the control loop.
- Applied multiprocessing on an ESP32 using both CPUs, improving processing speed and ensuring precise timing of tilt estimation at 1 kHz and PID calculation loop at 500 Hz.
- Decorruped IMU data by switching communication protocol from I2C to SPI, resisting motor power and signal noise and allowing for accurate usable sensor data.

Litter Collecting Drone Capstone, UBC

October 2025 – April 2026

- Rapidly prototyped various claw designs for collecting lightweight litter commonly found on hiking trails, 3D printing and assembling to test claw design effectiveness.
- Troubleshooted winch functionality using digital multimeter to monitor input and output signals, diagnosing the problem as mechanical rather than electrical.
- Retrofit claw and winch system onto drone circuit by redesigning drone connections and soldering.
- Crimped and soldered wires to splice a GPS module, allowing compatibility between brands of drone peripherals.

25:1 Gear Ratio Cycloidal Reducer, Personal Project

July 2025 – August 2025

- Maximized gearbox efficiency by choosing low friction materials and lubricating contact points, yielding an efficiency of 89%.
- Applied parametric equation in SOLIDWORKS to efficiently design cycloidal rotor, saving hours of time.
- Eliminated backlash by designing to 3D printer tolerances, producing tolerances of less than 0.02mm.
- Experimentally identified a skew error of 0.24% on the 3D printer axis and generated a python script to shift “gcode” instructions, eliminating mechanical skew error.

Quadruped Spider Robot, Personal Project

April 2025 – May 2025

- Designed and dimensioned components using SOLIDWORKS, aligning screw holes and minimizing material use.
 - Deployed inverse kinematics on spider legs with 2 degrees of freedom, optimizing smoothness of walking.
 - Prototyped circuit by designing a circuit schematic using KiCAD and testing on breadboard, minimizing risk.
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EDUCATION

University of British Columbia

Bachelor of Applied Science - Mechanical Engineering, Mechatronics Option

PROFESSIONAL AFFILIATIONS

EGBC - Engineers and Geoscientists British Columbia

2023 – Present